

# Section N

## Integration by Inspection

$$1. \int e^{5x} dx = \frac{1}{5} e^{5x} + C$$

$$2. \int x^8 dx = \frac{1}{9} x^9 + C$$

$$3. \int x^{-6} dx = -\frac{1}{5} x^{-5} + C$$

$$4. \int t^{\frac{5}{7}} dt = \frac{7}{12} t^{\frac{12}{7}} + C$$

$$5. \int x^{\frac{2}{3}} dx = \frac{3}{5} x^{\frac{5}{3}} + C$$

$$6. \int 4x^{-\frac{1}{2}} dx = (4)(2)x^{\frac{1}{2}} + C$$

$$7. \int \frac{1}{2} \theta^{-3} d\theta = \left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\theta^{-2} + C$$

$$8. \int x^{\frac{7}{2}} dx = \frac{2}{9} x^{\frac{9}{2}} + C$$

$$9. \int (x^3 - 2x + 7) dx = \left(\frac{1}{4}\right)x^4 - (2)\left(\frac{1}{2}\right)x^2 + 7x + C$$

$$10. \int (x^{-3} + x^{\frac{1}{2}} - 3x^{\frac{1}{4}} + x^2) dx \\ = -\frac{1}{2}x^{-2} + \frac{2}{3}x^{\frac{3}{2}} - (3)\left(\frac{4}{5}\right)x^{\frac{5}{4}} + \frac{1}{3}x^3 + C$$

$$11. \int (y^{\frac{2}{3}} - 4y^{\frac{1}{5}} + 4) dy \\ = \frac{3}{5}y^{\frac{5}{3}} - (4)\left(\frac{5}{6}\right)y^{\frac{6}{5}} + 4y + C$$

$$12. \int (7x^{\frac{3}{4}} - x^{\frac{1}{3}} + 4x^{\frac{1}{2}}) dx \\ = (7)\left(\frac{4}{7}\right)x^{\frac{7}{4}} - \frac{3}{4}x^{\frac{4}{3}} + (4)\left(\frac{2}{3}\right)x^{\frac{3}{2}} + C$$

$$13. \int (x + x^4) dx = \frac{1}{2}x^2 + \frac{1}{5}x^5 + C$$

$$14. \int (2 - u + 2u^2 - u^3) du \\ = 2u - \frac{1}{2}u^2 + (2)\left(\frac{1}{3}\right)u^3 - \frac{1}{4}u^4 + C$$

$$15. \int x^{\frac{1}{3}}(4 - 4x + x^2) dx \\ = \int \left(4x^{\frac{1}{3}} - 4x^{\frac{4}{3}} + x^{\frac{7}{3}}\right) dx \\ = (4)\left(\frac{3}{4}\right)x^{\frac{4}{3}} - (4)\left(\frac{3}{7}\right)x^{\frac{7}{3}} + \left(\frac{3}{10}\right)x^{\frac{10}{3}} + C$$

$$16. \int (t^{-2} - \cos t) dt = -t^{-1} - \sin t + C$$

$$17. \int (3\sin x - 4\cos x) dx \\ = -3\cos x - 4\sin x + C$$

$$18. \int e^{-x} dx = -e^{-x} + C$$

$$19. \int 2x e^{x^2} dx = e^{x^2} + C$$

$$20. \int \frac{1}{2x} dx = \frac{1}{2} \ln|x| + C$$

$$21. \int e^{3t} dt = \frac{1}{3} e^{3t} + C$$

$$22. \int \frac{3}{x} dx = 3 \ln|x| + C$$

$$23. \int \frac{1}{2x+1} dx = \frac{1}{2} \ln|2x+1| + C$$

$$24. \int \sinh 4x dx = \frac{1}{4} \cosh 4x + C$$

$$25. \int \sinh(4x+6) dx = \frac{1}{4} \cosh(4x+6) + C$$

$$26. \int \cosh(2x+3) dx = \frac{1}{2} \sinh(2x+3) + C$$

$$27. \int \tanh 3x dx = \int \frac{\sinh 3x}{\cosh 3x} dx \\ = \frac{1}{3} \ln|\cosh 3x| + C$$

$$28. \int \frac{x^2}{x^3-4} dx = \frac{1}{3} \ln|x^3-4| + C$$

$$29. \int \frac{5u^4}{u^5+1} du = \ln|u^5+1| + C$$

$$30. \int \left(1 + \frac{1}{t}\right) dt = t + \ln|t| + C$$

$$31. \int \frac{x^3}{x^2+1} dx \\ \text{Trick: } \int \frac{x(x^2+1-1)}{x^2+1} dx \\ = \int \frac{x(x^2+1)}{x^2+1} dx - \int \frac{x}{x^2+1} dx \\ = \int x dx - \int \frac{x}{x^2+1} dx \\ = \frac{1}{2}x^2 - \frac{1}{2} \ln|x^2+1| + C$$

OR Long division

$$\begin{array}{r} x \\ x^2+1 \overline{) x^3} \\ \underline{x^3+x} \phantom{0} \\ -x \phantom{0} \end{array}$$

$$\therefore \frac{x^3}{x^2+1} = x - \frac{x}{x^2+1}$$

$$\begin{aligned} \text{So } \int \frac{x^3}{x^2+1} dx &= \int \left( x - \frac{x}{x^2+1} \right) dx \\ &= \frac{1}{2} x^2 - \frac{1}{2} \ln |x^2+1| + C \end{aligned}$$

$$32. \int x^2 (2-x^3)^4 dx$$

$$= -\left(\frac{1}{3}\right)\left(\frac{1}{5}\right)(2-x^3)^5 + C$$

$$33. \int x e^{2x^2} dx = \frac{1}{4} e^{2x^2} + C$$

$$34. \int \sin \theta (\cos \theta - 3)^3 d\theta$$

$$= -\frac{1}{4} (\cos \theta - 3)^4 + C$$

$$35. \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = -2 \cos \sqrt{x} + C$$

$$36. \int \frac{\sin \theta}{1+\cos \theta} d\theta = -\ln |1+\cos \theta| + C$$

$$37. \int \tan \theta d\theta = \int \frac{\sin \theta}{\cos \theta} d\theta$$

$$= -\ln |\cos \theta| + C$$

$$38. \int \frac{1}{x \ln x} dx = \int \left(\frac{1}{x}\right) (\ln x) dx$$

$$= \int \frac{\frac{1}{x}}{\ln x} dx = \ln |\ln x| + C$$

$$39. \int \frac{\ln y}{y} dy = \frac{1}{2} (\ln y)^2 + C$$

$$40. \int (x+1) (x^2+2x)^{1/2} dx$$

$$= \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) (x^2+2x)^{3/2} + C$$

$$\text{Check: } \frac{d}{dx} \left( \left(\frac{1}{3}\right) (x^2+2x)^{3/2} \right)$$

$$= \left(\frac{1}{3}\right) \left(\frac{3}{2}\right) (x^2+2x)^{1/2} (2x+2)$$

$$= \left(2\right) \left(\frac{1}{2}\right) (x^2+2x)^{1/2} (x+1)$$